

# UD65640

## Application Note

### Orbit

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## 1. Target

Introducing the usage of UD65640 orbit.

## 2. Register:

### ORBIT: Shift pixel function(A3h)

|             |                                                   |              |                    |    |    |                  |    |                     |              |     |  |
|-------------|---------------------------------------------------|--------------|--------------------|----|----|------------------|----|---------------------|--------------|-----|--|
| A3h         | ORBIT                                             |              |                    |    |    |                  |    |                     |              |     |  |
|             | W / R                                             | B7           | B6                 | B5 | B4 | B3               | B2 | B1                  | B0           | Hex |  |
| Command     | —                                                 | 1            | 0                  | 1  | 0  | 0                | 0  | 1                   | 1            | A3h |  |
| parameter 1 | W / R                                             | 0            | 0                  | 0  | 0  | 0                | 0  | 0                   | ORBIT_E<br>N | 00h |  |
| parameter 2 | W / R                                             | ORBIT_V[3:0] |                    |    |    | ORBIT_H[3:0]     |    |                     |              | 00h |  |
| Description | ORBIT_EN<br>1: Orbit enable.<br>0: Orbit disable. |              |                    |    |    |                  |    |                     |              |     |  |
|             | ORBIT H:                                          |              |                    |    |    |                  |    |                     |              |     |  |
|             | ORBIT_H[3] = 1                                    |              |                    |    |    | ORBIT_H[3] = 0   |    |                     |              |     |  |
|             | ORBIT_H[2:0] = 0                                  |              | Center             |    |    | ORBIT_H[2:0] = 0 |    | Center              |              |     |  |
|             | ORBIT_H[2:0] = 1                                  |              | Left shift 1 pixel |    |    | ORBIT_H[2:0] = 1 |    | Right shift 1 pixel |              |     |  |
|             | ORBIT_H[2:0] = 2                                  |              | Left shift 2 pixel |    |    | ORBIT_H[2:0] = 2 |    | Right shift 2 pixel |              |     |  |
|             | ORBIT_H[2:0] = 3                                  |              | Left shift 3 pixel |    |    | ORBIT_H[2:0] = 3 |    | Right shift 3 pixel |              |     |  |
|             | ORBIT_H[2:0] = 4                                  |              | Left shift 4 pixel |    |    | ORBIT_H[2:0] = 4 |    | Right shift 4 pixel |              |     |  |
|             | ORBIT V:                                          |              |                    |    |    |                  |    |                     |              |     |  |
|             | ORBIT_V[3] = 1                                    |              |                    |    |    | ORBIT_V[3] = 0   |    |                     |              |     |  |
|             | ORBIT_V[2:0] = 0                                  |              | Center             |    |    | ORBIT_V[2:0] = 0 |    | Center              |              |     |  |
|             | ORBIT_V[2:0] = 1                                  |              | Up shift 1 pixel   |    |    | ORBIT_V[2:0] = 1 |    | Down shift 1 pixel  |              |     |  |
|             | ORBIT_V[2:0] = 2                                  |              | Up shift 2 pixel   |    |    | ORBIT_V[2:0] = 2 |    | Down shift 2 pixel  |              |     |  |
|             | ORBIT_V[2:0] = 3                                  |              | Up shift 3 pixel   |    |    | ORBIT_V[2:0] = 3 |    | Down shift 3 pixel  |              |     |  |
|             | ORBIT_V[2:0] = 4                                  |              | Up shift 4 pixel   |    |    | ORBIT_V[2:0] = 4 |    | Down shift 4 pixel  |              |     |  |
|             | ORBIT_V[2:0] = 5                                  |              | Up shift 5 pixel   |    |    | ORBIT_V[2:0] = 5 |    | Down shift 5 pixel  |              |     |  |
|             | ORBIT_V[2:0] = 6                                  |              | Up shift 6 pixel   |    |    | ORBIT_V[2:0] = 6 |    | Down shift 6 pixel  |              |     |  |
|             | 1 = Must fill 1; 0 = Must fill 0; X = Don't care. |              |                    |    |    |                  |    |                     |              |     |  |
|             | Restriction                                       |              |                    |    |    |                  |    |                     |              |     |  |

### 3. Flow

#### 3.1 Orbit Flow

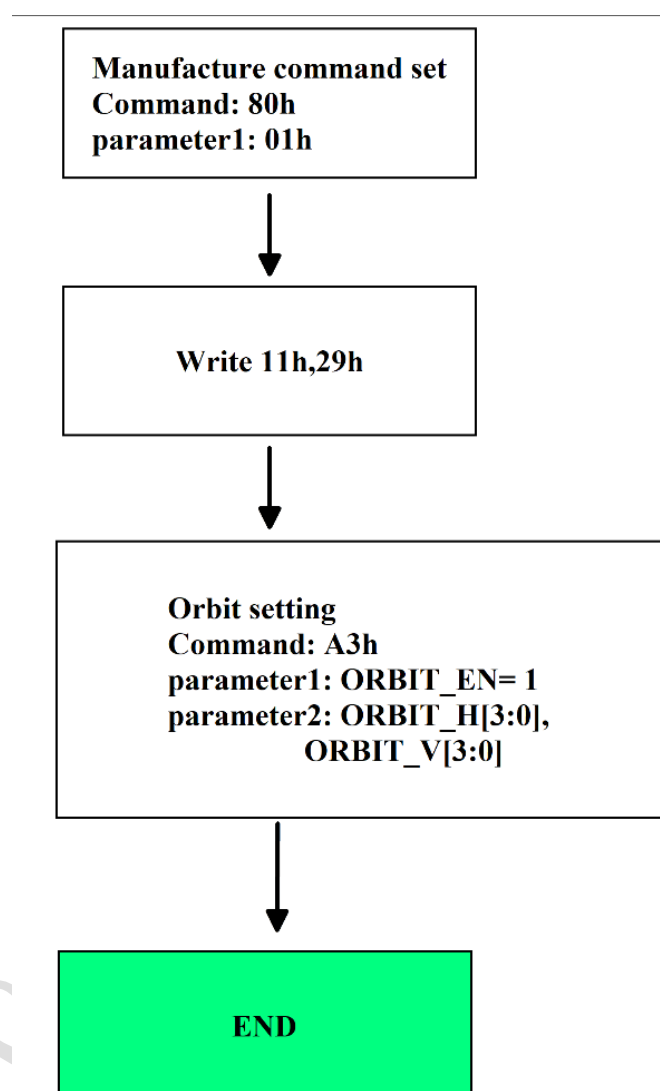


Figure 1 – Orbit Flow

## 4. Example

### 4.1 Orbit Example

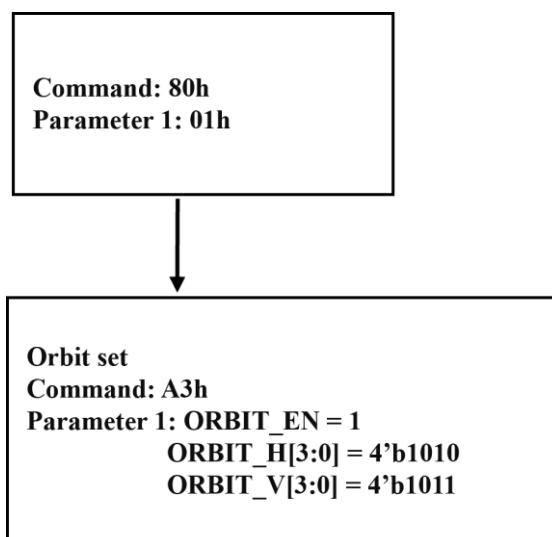


Figure 2 – Orbit Example

Orbit before

Orbit after

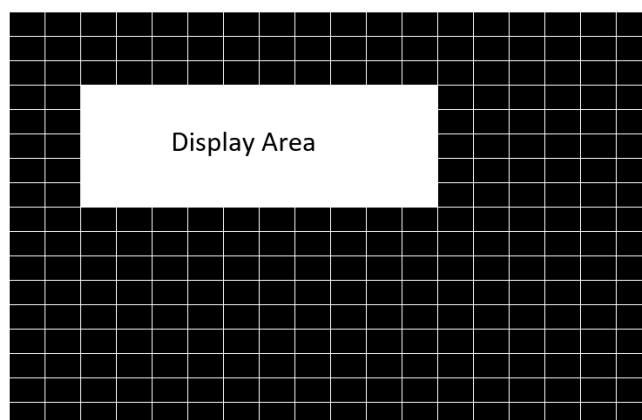
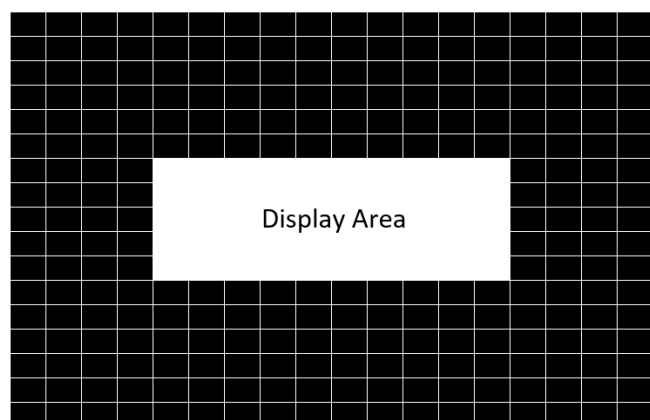


Figure 3 – Orbit Display

## 5. Revision History

| Version No.     | Date     | Page | Description     |
|-----------------|----------|------|-----------------|
| AN-K0002-240704 | 2024/7/4 | 6    | Initial Version |